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Instructor: Dr Khursheed Ali

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Submission Profile

Name: Muhammad Anas Qadri

CMS ID: 053-24-0016

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GitHub repository URL

<https://github.com/anasqadri-dev/software-quality-engineering>

docs/ep-analysis.md with the equivalence-class tables for all three inputs and a short note on EP's boundary-blind-spot limitation

<https://github.com/anasqadri-dev/software-quality-engineering/blob/main/LABs/LAB%205/sqe-gradebook/docs/ep-analysis.md>

Equivalence Partitioning Analysis — GradeBook

1. Introduction

This document contains the Equivalence Partitioning (EP) analysis for the GradeBook system.

Equivalence Partitioning divides an input domain into classes where the system is expected to behave similarly. One representative value is selected from each class to reduce redundant testing while maintaining useful defect-detection coverage.

2. Input 1 — Student Score

The score domain is divided into valid and invalid equivalence classes.

Class ID	Equivalence Class	Input Range	Representative Value	Expected Result
SC-1	Invalid low	score < 0	-10	Reject with error
SC-2	F grade	0-59	45	Return F
SC-3	D grade	60-69	65	Return D
SC-4	C grade	70-79	75	Return C
SC-5	B grade	80-89	85	Return B
SC-6	A grade	90-100	95	Return A
SC-7	Invalid high	score > 100	110	Reject with error

3. Input 2 — Number of Scores per Student

The business rule requires a student to have between 1 and 5 scores.

Class ID	Equivalence Class	Input Range	Representative Value	Expected Result
NS-1	Invalid low	0 scores	0	Reject with error
NS-2	Valid	1-5 scores	3	Accepted
NS-3	Invalid high	6 or more scores	6	Reject with error

4. Input 3 — Student Name

The student name must be non-empty (begin with a nonempty length of 50 characters) and may contain only letters, spaces, and hyphens.

Class ID	Equivalence Class	Input Example	Expected Result
NM-1	Valid typical name	Ali Khan	Accepted
NM-2	Empty string	"	Reject with error
NM-3	Over-length name	More than 50 characters	Reject with error
NM-4	Digits or symbols	Ali123	Reject with error

5. Limitation of Equivalence Partitioning

Equivalence Partitioning does not take in exact boundary values. Therefore, it can miss off-by-one defects at the edges of equivalence classes.

For example, score classes contain boundaries such as 59/60, 69/70, 79/80, 89/90, and 99/100. EP selects a representative from each class but does not test every boundary.

Boundary Value Analysis will be used in a later EP to specifically target these edge values.

7. Full EP-Test Execution Summary

The complete EP test suite was executed after implementing the score, name, and name-validation functionality.

Test Summary

Test Area	Result
Letter Grade EP Tests	PASS — 1/1
Score Count EP Tests	PASS — 1/1
Name Validation EP Tests	FAIL — 4/4
Total Test Suite	FAIL

Name Validation Results

Class	Representation	Result
Valid typical name	"Ali Khan"	PASS
Empty string	"	FAIL
Over-length name	51 characters	FAIL
Characters/digits	"Ali123"	FAIL

Overall Result

All implemented Equivalence Partitioning tests passed successfully. The test suite covers the defined equivalence classes for letter grades, score counts, and student names.

Note: Equivalence Partitioning is not intended to replace boundary value analysis. Boundary values such as the exact 50-character name limit and score/grade boundaries require additional boundary-focused testing.

tests/LetterGradeTest.java, tests/RosterTest.java,
tests/NameValidation.java — all passing

LetterGrade Test

```
Test Runner for Java
✓ LetterGradeTest $(symbol-namespace) <Def...
✓ testAGradeClass() $(symbol-class) LetterGra...
✓ testBGradeClass() $(symbol-class) LetterGra...
✓ testCGradeClass() $(symbol-class) LetterGra...
✓ testDGradeClass() $(symbol-class) LetterGra...
✓ testFGradeClass() $(symbol-class) LetterGra...
✓ testInvalidHighScoreClass() $(symbol-class...)
✓ testInvalidLowScoreClass() $(symbol-class) ...
```

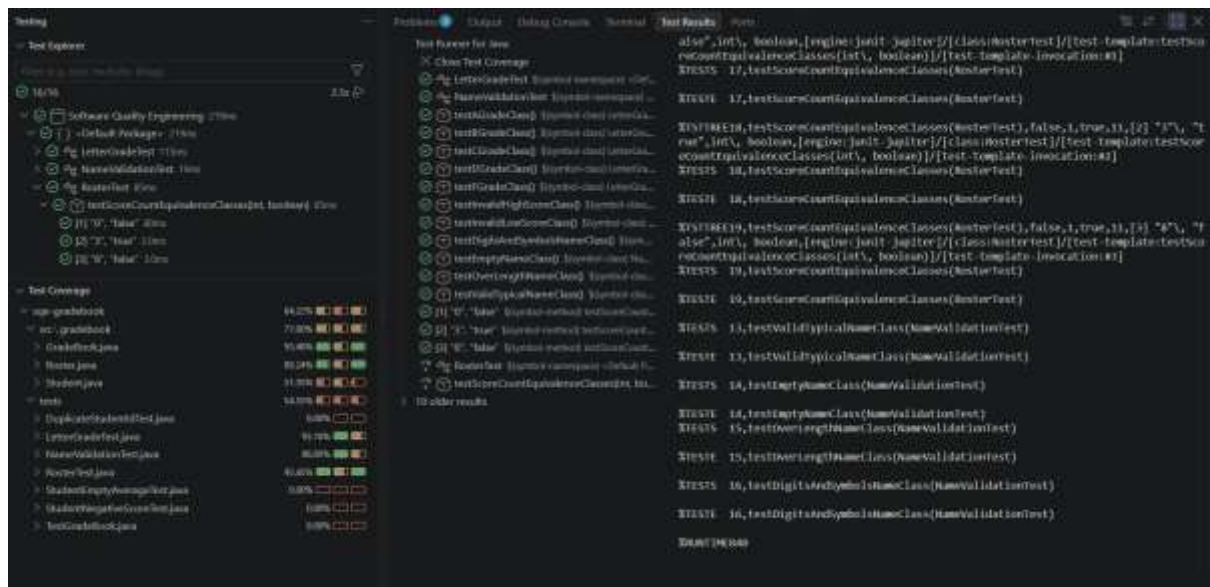
NameValidation Test

```
Test Runner for Java
✓ NameValidationTest $(symbol-namespace) ...
✓ testDigitsAndSymbolsNameClass() $(sym...
✓ testEmptyNameClass() $(symbol-class) Na...
✓ testOverLengthNameClass() $(symbol-clas...
✓ testValidTypicalNameClass() $(symbol-clas...
```

Roster Test

```
Test Runner for Java
✓ [1] "0", "false" $(symbol-method) testScoreCount..
✓ [2] "3", "true" $(symbol-method) testScoreCount...
✓ [3] "8", "false" $(symbol-method) testScoreCount..
```

Junit run summary (terminal output) showing all EP tests green



THE END